Abhijeet Rathore

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Machine Learning Researcher, Time-Series and Solar Energy Forecasting

Summary

Machine learning researcher (PhD, IIT Roorkee) specialising in time-series forecasting for solar energy. First author in *Renewable Energy* (Elsevier, 2025) and co-inventor on a granted Indian patent for ML-based solar PV power forecasting. Builds end-to-end ML systems: data pipelines, deep learning and gradient-boosting models, probabilistic forecasts, REST APIs and deployed web apps. Current research fine-tunes large language models (LLMs) for time-series forecasting with LoRA and QLoRA, adds physical constraints to them (physics-informed ML), and compresses models through knowledge distillation to run on edge devices such as the Raspberry Pi.

Education

Indian Institute of Technology (IIT) Roorkee — Roorkee, India
PhD, Dept. of Hydro and Renewable Energy; CGPA: 9.415 — *July 2024 – Present*

Energy Institute, Bengaluru (Centre of RGIPT) — Bengaluru, India
M.Tech, Power and Energy Systems Engineering; CPI: 9.15; Institute Gold Medal — *Aug 2020 – June 2022*

SRMCEM (Affiliated to Dr. A.P.J. Abdul Kalam Technical University) — Lucknow, India
B.Tech, Electrical and Electronics Engineering; 77.48% — *July 2013 – June 2017*

CBSE: 10+2 (80%), 2013; High School (CGPA 8.2), 2011 — Lucknow, India

Experience

Doctoral Researcher — IIT Roorkee
July 2024 – Present — *Roorkee, India*

- Fine-tuning open large language models (Gemma, Llama, Qwen) for multivariate time-series forecasting of solar power using parameter-efficient fine-tuning (LoRA, QLoRA) in PyTorch and Hugging Face, with physics-based constraints so forecasts stay physically consistent; local inference with llama.cpp.
- Building mixture-of-experts forecasting systems with model routing and blending to combine specialised models.
- Deploying lightweight ML models on edge devices (Raspberry Pi) with sensor integration, using knowledge distillation to shrink larger models for on-device inference.
- Built India Renewables Forecast, a live web app giving 7-day solar PV and wind power forecasts, with satellite-corrected first hours, for any location in India (app.abhijeetrathore.in); see Projects.

Research Engineer — IIT Roorkee
July 2022 – July 2024 — *Roorkee, India*

- Developed a day-ahead solar irradiance forecasting model (NA-MEMD signal decomposition, FFT-based frequency grouping, attention-based LSTM) that cut mean RMSE by 36.9% against an attention-LSTM baseline (R = 0.972, nRMSE 6.33%) and training time by 58.8% across four Indian sites. Published in *Renewable Energy* (Elsevier) and granted Indian Patent No. 580682.
- Managed data from a six-station weather and PV monitoring network across North India, owning the end-to-end workflow: acquisition, cleaning, quality control, feature engineering and model training.
- Automated data collection and preprocessing pipelines for all six sites, reducing manual effort by over 90%.
- Led backend development and model integration for ML-Sol, a commercial solar forecasting web service (ml-sol.com); built Flask and Streamlit interfaces so non-specialists could run forecasts.
- Explained models and results to technical and non-technical stakeholders.

Data Analyst — Del2Infinity Energy Consulting Pvt. Ltd.
March 2018 – July 2020 — *Remote*

- Analysed solar plant performance data for plant owners using Python and Excel, building statistical and numerical models to support energy-management decisions.
- Applied statistical testing (ANOVA) to compare plant performance and identify significant drivers of generation.
- Collected, cleaned and validated data, and turned the analysis into clear reports and recommendations for clients.

Technical Skills

Languages and tools: Python, MATLAB, JavaScript (React), Excel.
Machine learning: Time-series forecasting, deep learning (LSTM, attention mechanisms), gradient boosting (LightGBM), physics-informed machine learning, mixture of experts (model routing, ensemble blending), probabilistic forecasting (quantile regression, prediction intervals), feature engineering, hyperparameter tuning, model evaluation and out-of-sample validation, signal decomposition (EMD, FFT), anomaly detection, statistical analysis (ANOVA).
LLMs and generative AI: Fine-tuning large language models (Gemma, Llama, Qwen), parameter-efficient fine-tuning (PEFT: LoRA, QLoRA), prompt engineering, LLMs for time-series forecasting, llama.cpp for local inference.
Edge AI: Raspberry Pi, sensor integration, IoT data acquisition, lightweight ML models, knowledge distillation, model compression, on-device inference.

Libraries: PyTorch, Hugging Face (Transformers, PEFT), TensorFlow, Keras, Keras Tuner, scikit-learn, LightGBM, pandas, NumPy, Matplotlib, pvlib.
Deployment and data engineering: FastAPI, Flask, Streamlit, REST APIs, ETL and data pipelines, CI/CD with GitHub Actions, Git, Vercel, Render.
Energy domain: Solar irradiance and PV power forecasting, numerical weather prediction (NWP), satellite irradiance data (NASA POWER, NSRDB, Meteosat), satellite nowcasting, clear-sky modelling, PV yield modelling, wind power curves, grid integration, power system simulation (MATLAB/Simulink).

Projects

- **India Renewables Forecast (live web app, Jan – Mar 2026; app.abhijeetrathore.in):** Built end to end: data, model, API and front end. One India-wide LightGBM model corrects numerical weather prediction irradiance using satellite data, with P10/P90 quantile models for an 80% prediction interval; pvlib PV modelling and hub-height wind power curves convert it to power. Trained on 2022–2023 and tested on 2024: hourly irradiance RMSE fell from 101 to 78 W/m^2 (23% better than the raw forecast) and improved 10% at 8 sites held out of training. Added a satellite nowcast: the latest Meteosat-derived irradiance corrects the first forecast hours with a lead-time-decaying weight, cutting error a further 3.2% in the first live hour at held-out sites. Converts forecasts to rupees: deviation charges under Indian scheduling rules and project NPV, IRR and LCOE with sourced regulatory defaults. Stack: Python, LightGBM, pvlib, FastAPI, React, GitHub Actions, Render, Vercel. [case study]
- **M.Tech Thesis: Yield Modelling and Data-Driven Feasibility Analysis of Solar PV and Solar Thermal Systems:** Data-driven solar PV energy yield model with integrated optimisation for site-specific feasibility and parameter tuning of utility-scale and distributed installations (MATLAB, statistical modelling).
- **B.Tech Project: Enhancement of Voltage Quality in Isolated Power Systems:** Battery-backed series compensator in MATLAB/Simulink to reduce total harmonic distortion (THD) and improve voltage stability in isolated grids.

Patent and Publications

- R. Singh, R. Sharma, **A. Rathore**, and P. Gupta, "A System and Method for Forecasting Solar Photovoltaic Power Using Machine Learning," Indian Patent No. 580682, granted Feb. 16, 2026.
- **A. Rathore**, et al., "Day ahead solar forecast using long short term memory network augmented with Fast Fourier transform-assisted decomposition technique," *Renewable Energy* 247 (2025): 123021. doi:10.1016/j.renene.2025.123021
- **A. Rathore**, et al., "Energy, exergy and performance analysis of a 380 kWp roof-top PV plant assisted with data-driven models for energy generation," *Journal of Thermal Engineering* 10.5 (2024): 1164–1183.
- **A. Rathore**, et al., "An IoT-Based Automatic Light Control System for Smart Buildings," in *International Conference on Signal, Machines, Automation, and Algorithm*, Springer Nature Singapore, 2023. doi:10.1007/978-981-97-6349-8_42
- Book chapter: "Configurations of Converters applied to Electric Vehicles with comprehensive control strategies and optimization," Chapter 13 in *Power Electronics for Electric Vehicles and Energy Storage: Emerging Technologies and Developments*, CRC Press (Taylor & Francis), 2023.
- **A. Rathore** and S. Sundaram, "Building Energy Management Through Load Control Using Internet of Things," in *2021 Innovations in Power and Advanced Computing Technologies (i-PACT)*, 2021, pp. 1–6. doi:10.1109/i-PACT52855.2021.9696596

Courses and Certifications

- Generative AI and LLMs: Architecture and Data Preparation; Gen AI Foundational Models for NLP and Language Understanding (IBM, Coursera).
- Neural Networks and Deep Learning (DeepLearning.AI, Coursera); Intro and Intermediate Machine Learning (Kaggle); Machine Learning and Data Science Bootcamp (Udemy).
- Google Data Analytics Certificate (Google, Coursera); Programming for Everybody and Python Data Structures (University of Michigan, Coursera).
- MathWorks: MATLAB and Simulink Onramp, Machine Learning with MATLAB, Solving ODEs with MATLAB.

Industrial Training

- UPPCL, Lucknow: 132 kV substation operations, transformers, switchgear, CT/PT and grid protection.
- Industrial Automation workshop (Sofcon): PLC programming, HMI, SCADA. UNICON PLC Automation: MATLAB control systems and signal processing.

Honors and Awards

- **Institute Gold Medal**, Rajiv Gandhi Institute of Petroleum Technology (RGIPT): best academic performance, M.Tech Power and Energy Systems Engineering.
- **Certificate of Merit**, SRMCEM: outstanding academic performance in the 2nd, 3rd and 4th years of B.Tech.
- **Merit Certificate**, CBSE: Class 12 examinations.

References

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